

Lab 6 & 7 - Report

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Group: Mon 11:00-13:00

Lab 6

1. Part 1: Examine Network Requirements

Step 1.

a) $32 - 25 = 7$

$2^7 - 2 = 126$

2) $40 + 25 + 5 + 4 + 4 + 2 = 80$

3) 6 subnets

Step 2

BR1 LAN

$40 < 2^6$ $32 - 6 = 26$

$2^6 - 2 = 62$

Step 3

BR2 LAN

$25 < 2^5$ $32 - 5 = 27$

$2^5 - 2 = 30$

Step 4

4, 5 < 2^3 $32 - 3 = 29$

$2^3 - 2 = 6$

Step 5

$2 < 2^2$ $32 - 2 = 30$

$2^2 - 2 = 2$

Step 1: Determine how many host addresses and subnets are available.

How many host addresses are available in a /25 network?

126

What is the total number of host addresses needed in the topology diagram?

80

How many subnets are needed in the network topology?

6

Step 2: Determine the largest subnet.

What is the subnet description (e.g. BR1 LAN or BR1-BR2 link)?

BR1 LAN

How many IP addresses are required in the largest subnet?

40

What subnet mask can support that many host addresses?

/26 (255.255.255.192)

How many total host addresses can that subnet mask support?

62

Can you subnet the 192.168.33.128/25 network address to support this subnet?

Yes

What are the network addresses that would result from this subnetting?

192.168.33.128/26 and 192.168.33.192/26

Step 3: Determine the second largest subnet.

What is the subnet description?

BR2 LAN

How many IP addresses are required for the second largest subnet?

25

What subnet mask can support that many host addresses?

/27 or 255.255.255.224

How many total host addresses can that subnet mask support?

30

Can you subnet the remaining subnet again and still support this subnet?

Yes

What are the network addresses that would result from this subnetting?
192.168.33.192/27 and 192.168.33.224/27

Step 4: Determine the third largest subnet.

What is the subnet description?

BR2 IoT LAN

How many IP addresses are required for the next largest subnet?

5

What subnet mask can support that many host addresses? /29 or
255.255.255.248

How many total host addresses can that subnet mask support?

6

Can you subnet the remaining subnet again and still support this subnet?
yes

What are the network addresses that would result from this subnetting?
192.168.33.224/29, 192.168.33.232/29, 192.168.33.240/29, 192.168.33.248/29

Step 5: Determine the fourth largest subnet.

What is the subnet description?

BR11-BR2 Link

How many IP addresses are required for the next largest subnet?

2

What subnet mask can support that many host addresses?
/30 or 255.255.255.252

How many total host addresses can that subnet mask support?

2

Can you subnet the remaining subnet again and still support this subnet?
yes

What are the network addresses that would result from this subnetting?
192.168.33.248/30, 192.168.33.252/30

2. Part 1: Examine Network Requirements

Part 2

1) BR1 LAN (40, 2^6)

$$32 - 6 = 26$$

$$128 + 1 = 129$$

$$128 + 64 - 1 = 191$$

2) BR2 ~~LAN~~ (25, 2^5)

$$32 - 5 = 27$$

$$192 + 1 = 193$$

$$192 + 32 - 1 = 223$$

3) BR2 ~~LAN~~ IoT (5, 2^3)

$$32 - 3 = 29$$

$$224 + 1 = 225$$

$$224 + 8 - 1 = 231$$

4) BR2 CCTV (4, 2^3)

$$32 - 3 = 29$$

$$232 + 1 = 233$$

$$232 + 8 - 1 = 239$$

5) BR2 HVAC (4, 2^3)

$$32 - 3 = 29; \quad 240 + 1 = 241; \quad 240 + 8 - 1 = 247$$

6) BR1 - BR2 Link (2, 2^4)

$$32 - 2 = 30; \quad 248 + 1 = 249$$

$$248 + 4 - 1 = 251$$

Step 1: Calculate the subnet information.

Subnet Description	Number of Hosts Needed	Network Address /CIDR	First Host Address	Broadcast Address
BR1 LAN	40	192.168.33.128/25	192.168.33.129	192.168.33.191
BR2 LAN	25	192.168.33.192/27	192.168.33.193	192.168.33.223
BR2 IoT LAN	5	192.168.33.224/29	192.168.33.225	192.168.33.231
BR2 CCTV LAN	4	192.168.33.232/29	192.168.33.233	192.168.33.239
BR2 HVAC C2LAN	4	192.168.33.240/29	192.168.33.241	192.168.33.247
BR1-BR2 Link	2	192.168.33.248/30	192.168.33.249	192.168.33.251

Step 2: Complete the device interface address table.

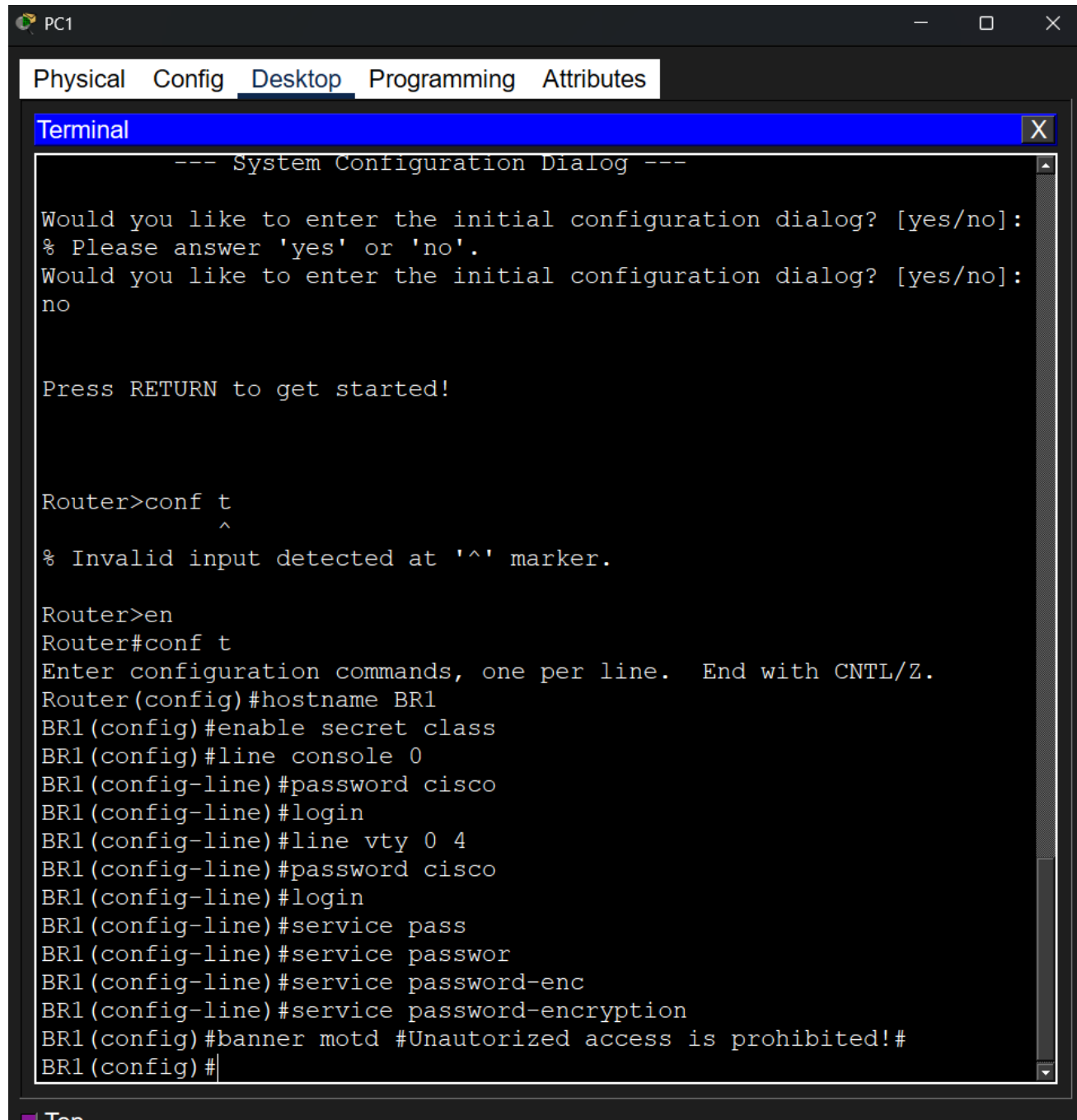
Device	Interface	IP Address	Subnet Mask	Device Interface
BR1	G0/0/0	192.168.33.249	255.255.255.252	BR1-BR2 Link
	G0/0/1	192.168.33.129	255.255.255.192	40 Host LAN
BR2	G0/0/0	192.168.33.250	255.255.255.252	BR1-BR2 Link
	G0/0/1	192.168.33.193	255.255.255.224	25 Host LAN

3. Part 3: Cable and Configure the IPv4 Network

Step 1



Step 2



The screenshot shows a terminal window titled 'PC1' with tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Terminal' tab is active, displaying a 'System Configuration Dialog'. The user is prompted to enter the initial configuration dialog, which they decline. They then enter 'conf t' (which is rejected as invalid) and 'en' to enter configuration mode. In configuration mode, they set the hostname to 'BR1', enable secret class, configure console and vty lines with password 'cisco' and login, set service password encryption, and set a MOTD banner. The terminal text is as follows:

```
--- System Configuration Dialog ---

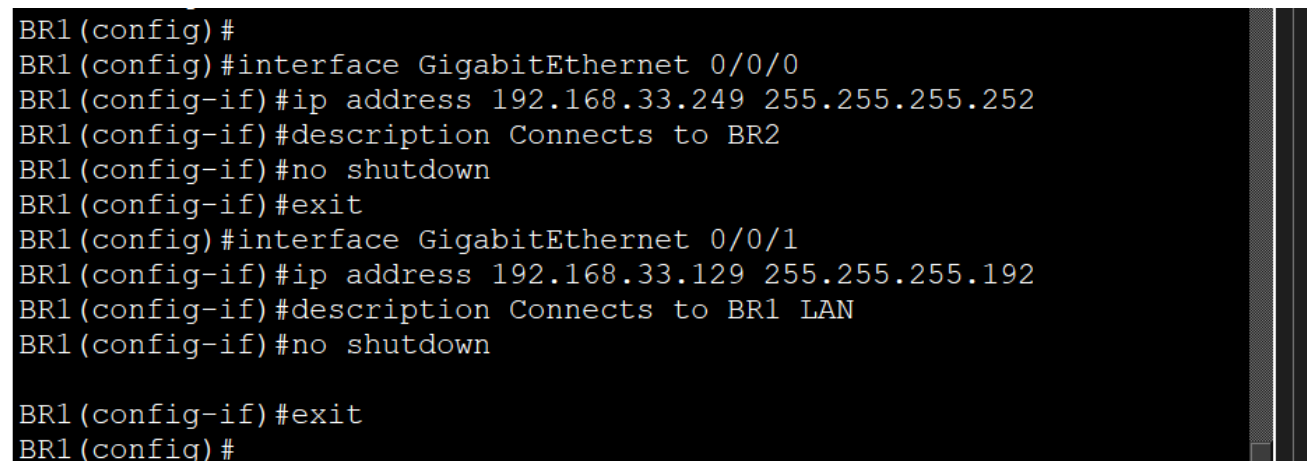
Would you like to enter the initial configuration dialog? [yes/no]:
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]:
no

Press RETURN to get started!

Router>conf t
      ^
% Invalid input detected at '^' marker.

Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname BR1
BR1(config)#enable secret class
BR1(config)#line console 0
BR1(config-line)#password cisco
BR1(config-line)#login
BR1(config-line)#line vty 0 4
BR1(config-line)#password cisco
BR1(config-line)#login
BR1(config-line)#service pass
BR1(config-line)#service passwor
BR1(config-line)#service password-enc
BR1(config-line)#service password-encryption
BR1(config)#banner motd #Unauthorized access is prohibited!#
BR1(config)#
```

Step 3



The screenshot shows a terminal window with the following configuration commands for interface GigabitEthernet 0/0/0 and 0/0/1 on router BR1:

```
BR1(config)#
BR1(config)#interface GigabitEthernet 0/0/0
BR1(config-if)#ip address 192.168.33.249 255.255.255.252
BR1(config-if)#description Connects to BR2
BR1(config-if)#no shutdown
BR1(config-if)#exit
BR1(config)#interface GigabitEthernet 0/0/1
BR1(config-if)#ip address 192.168.33.129 255.255.255.192
BR1(config-if)#description Connects to BR1 LAN
BR1(config-if)#no shutdown

BR1(config-if)#exit
BR1(config)#
```

Step 4

```
BR1#wr mem
Building configuration...
[OK]
BR1#
```

Step 5

```
BR2#
BR2#ping 192.168.33.249

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.33.249, timeout is 2
seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1
ms

BR2#
```

```
BR1#ping 192.168.33.250

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.33.250, timeout is 2
seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/3 ms

BR1#
```

Reflection Question

Can you think of a shortcut for calculating the network addresses of consecutive /30 subnets?

The step between network addresses in a /30 mask is always 4. To find the next /30 subnet, simply add 4 to the current network address.